

MODEL CARD

Student Dropout Predictor — Version 3.0

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Model Details

Model name	Student Dropout Predictor
Version	3.0 (leakage-free, binary)
Model type	Random Forest Classifier (class-weighted)
Developer	Abdullahi Isah, Data Analyst
Date	September 2026
Tools	Python (Pandas, Scikit-learn, Matplotlib)
Status	Proof-of-concept — not yet deployed

Intended Use

The model is intended to serve as an early-warning system that flags students at risk of dropping out, using information available to a school at the start of a semester. It is designed to inform counselor conversations and targeted support — not to trigger any automatic administrative action.

This version was trained on higher-education data and is treated as a proof-of-concept for the methodology (data → features → model → early-warning output). The intended future application is Kano State primary and secondary (basic education) students, where dropout is more prevalent and more consequential, using locally sourced data and a substantially rebuilt feature set.

Training Data

Public Kaggle dataset of 4,424 higher-education student records (Portuguese institution). Class distribution: 1,421 students (32%) dropped out; 2,209 (50%) graduated; 794 (18%) remained enrolled. For this version, the target was collapsed into a binary outcome:

- Dropout — 1,421 students
- Not Dropout (Graduate + Enrolled) — 3,003 students

Split: 80% train / 20% test.

Features

28 features, restricted to information available at the start of the semester. All second-semester features were removed in this version to eliminate data leakage present in earlier iterations.

Feature categories

- Demographics — age, gender, marital status
- Family background — parents' education, occupation
- Financial status — debtor status, tuition up to date, scholarship holder
- First-semester academic performance
- Engineered features — total approved units, approval rate, average grade, financial risk indicator

Top predictive features

- Tuition fees up to date
- Scholarship holder
- Curricular units approved
- Age at enrollment
- Debtor status

Performance

Metric	v2 (3-class, leaky)	v3 (binary, leakage-free)
Overall accuracy	77.51%	84.18%
Dropout recall	76%	66%
Dropout precision	—	86%
Not Dropout recall	—	94%

The v2 dropout recall of 76% relied on second-semester features that would not be available at prediction time and is therefore considered unreliable. The v3 figures are leakage-free and reflect realistic deployment conditions.

In practical terms: of 100 students actually at risk of dropping out, the model correctly flags 66 — a six-fold improvement over no early-warning system.

Class weights from 1 to 10 were tested to raise recall; recall remained at approximately 65–66% across this range, indicating the current feature set has reached its predictive ceiling. Further recall gains will require richer data (e.g. attendance, household factors, distance to school) rather than further tuning.

Limitations

- Does not predict with certainty; recall is capped at ~66% given current features.
- Trained on Portuguese higher-education data; features and relationships may not transfer to Kano's basic-education context (e.g. household poverty, distance to school, early marriage, seasonal labor are not represented in this dataset).
- Not yet tested for gender bias — subgroup performance by gender must be reviewed before any deployment.
- Does not replace human judgment and should not be used to trigger automatic action.

Ethical Considerations & Safeguards

- Human-in-the-loop: model output informs counselor conversations only, never automatic action.
- Framing: students are flagged as “needing support,” never labeled as “likely to fail.”
- Every flagged case must be reviewed by a human before any action is taken.
- Gender-disaggregated performance will be reviewed before deployment.
- Student data consent and privacy norms must be established prior to use of any real student data.
- Recommended rollout: pilot in one district before any wider deployment.

Recommendations for Next Version

- Source local, anonymized student-level data from the Kano State Ministry of Education (pilot LGA).
- Rebuild the feature set around basic-education risk factors (attendance, household income proxy, distance to school, sibling enrollment).
- Conduct gender-fairness evaluation before any deployment.
- Build a simple dashboard for school administrators once retrained on local data.

Contact & Maintenance

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